

Karan Shihire

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EDUCATION

Modern College of Arts, Science and Commerce

Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0

Pune, Maharashtra

Jul 2023 – May 2026

CERTIFICATIONS

Cisco Certified Network Associate (CCNA)

Routing & Switching, TCP/IP, VLANs, Subnetting, Network Security Fundamentals

Cisco

Oracle Cloud Infrastructure Foundations Associate

Cloud Architecture, OCI Core Services, Identity & Security

Oracle

PROJECTS

- **AI-Powered Network Intrusion Detection System (IDS)** | *Python, Scapy, Isolation Forest* [GitHub](#)
 - **Network layer packet capture:** Used Scapy to sniff live IP traffic, parsing protocol headers (TCP/UDP/ICMP) and computing per-IP **sliding-window packet rate** (10-second window) as a real-time traffic feature.
 - **ML anomaly detection:** Trained an **Isolation Forest** model on **5,000+ live-captured packets** across features (packet rate, packet length, protocol) for unsupervised detection of abnormal network behavior — no labelled attack data required.
 - **Multi-rule detection engine:** Designed 5 rule layers mapped to real attack patterns — **DoS/flood** (rate > 20 p/s), **Port Scan** (small packets at elevated rate), **ICMP abuse** (oversized ping), and **ML anomaly** (Isolation Forest output = -1).
 - **End-to-end pipeline:** Packet capture → feature extraction → ML inference → structured alert logging, running concurrently on live traffic with per-packet threat classification (NORMAL / ALERT).
 - **Operational UI:** Built a Flask + Chart.js dashboard to visualize the live detection pipeline — rolling packet-rate graph, per-packet color classification, and top-attacker panel — making system output human-readable.
- **Network Monitoring System** | *Python, ICMP, Socket Programming, Flask* [GitHub](#)
 - **ICMP-level monitoring:** Worked directly at the network layer using raw ICMP packets to measure round-trip latency, detect packet loss, and identify unreachable hosts across multiple devices.
 - **Failure detection:** Built structured downtime logging to record and alert on network outages in real time — tracking availability trends across monitored subnets.
 - Developed a Flask dashboard as a supporting interface to display device status and uptime history.
- **QuizNet — AI-Powered Quiz Generator** | *Python, Django, Google Gemini API, SQLite* [GitHub](#)
 - Integrated Google Gemini LLM API to dynamically generate MCQ, reasoning, and true/false questions — demonstrates practical LLM API usage and prompt-driven AI output handling.
 - Built Django backend with session management, scoring logic, and a clean UI to present AI-generated content effectively.

TECHNICAL SKILLS

Networking: TCP/IP, ICMP, DNS, DHCP, OSI Model, Routing & Switching, VLANs, Subnetting, Packet Analysis (Wireshark, Scapy)

Programming: Python (primary), Java, SQL

Tools: Scapy, Wireshark, Cisco Packet Tracer, Git, GitHub, VS Code, Cursor

Web: Flask, Django, Chart.js — used to build dashboards and UIs for AI and network projects

Cloud: Oracle OCI, Render, Netlify